

QUICK ACTIVITY 2: CANTEEN GROUP ORDER

A canteen sells rice meals at ₱85 each and drinks at ₱25 each. Every order receives a 10% discount on the rice-meal subtotal only. The canteen also charges a ₱5 packaging fee per rice meal. Develop an algorithm that calculates the final bill and the customer's change.

DECOMPOSITION

Input	Process	Output
Read rice meal amount, drink amount, and the payment amount	Calculate rice meal amount, drink amount, discount on rice meals, packaging fee, discounted meal amount, total bill, change	Present total bill and change rounded to 2 decimal places

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START

 READ quantity, payment

 riceMeals_amount $\leftarrow$ quantity $\times$ 85

 drinks_amount $\leftarrow$ quantity $\times$ 25

 discountedAmount $\leftarrow$ riceMeals_amount $\times$ 0.10

 printingFee $\leftarrow$ quantity $\times$ 5

 total_amount $\leftarrow$ discountedAmount + printingFee
 change $\leftarrow$ payment - totalAmount

 DISPLAY totalBill, change

END

DRY RUN CALCULATIONS

Test Case 1

Inputs:

Rice Meals = 3

Drinks = 4

Payment = ₱500

Steps	Variable	Calculations	Solution
1	Rice Meals Amount	3×85	₱225.00
2	Drinks Amount	4×25	₱ 100.00
3	Discount	225×0.10	₱ 25.50
4	Packaging Fee	3×5	₱ 15.00
5	Final Bill	$255 - 25.50 + 100 + 15$	₱344.50
6	Change	$500 - 344.50$	₱155.50

Output:

Total Bill = ₱344.50

Change = ₱155.50

Test Case 2

Inputs:

Rice Meals = 5

Drinks = 3

Payment = ₱500.00

Steps	Variable	Calculation	Solution
1	Rice Meals Amount	$5 \times \text{₱}85.00$	₱425.00
2	Drinks Amount	$3 \times \text{₱}15.00$	₱ 75.00
3	Discounted Amount	$\text{₱}425 \times 0.10$	₱42.50
4	Packing Fee	5×5	₱25.00
5	Final Bill	$425 - 42.50 + 75 + 25$	₱482.50
6	Change	$\text{₱ } 600 - \text{₱}482.50$	₱ 117.50

Output:

Final Bill = ₱482.50

Change = ₱ 111.50

Explain: Why would applying the 10% discount to the entire bill produce an incorrect result?

- If the 10% will be applied to the entire bill, it will be incorrect because the rules clearly stated that the discount should be applied to rice meals only. If it is added to the total bill, the answer would probably decrease and may result in unfairness on the canteen owner, while giving the customer an extra discount.